

COURSE CODE	COURSE TITLE	L	T	P	TW & SL	TH	C
UCS3513	SYSTEM DESIGN LABORATORY	0	0	45	0	45	1.5

OBJECTIVES

- Build proficiency in evaluating tradeoffs between different system design approaches
- Gain practical experience designing scalable, reliable, and maintainable systems
- Apply theoretical concepts to realistic system design scenarios
- Prepare students for system design interviews and real-world engineering challenges

SUGGESTED EXPERIMENTS

1. Design a system that supports zero to millions of users.
2. Design and develop a rate limiter which is used to control the rate of traffic sent by a client or a service.
3. Design and develop a Consistent hashing to distribute requests/data efficiently and evenly across servers.
4. Design and develop a Key-Value Store to identify the value uniquely by a key.
5. Design and develop a unique ID generator with auto-increment feature for primary key in Distributed Systems.
6. Design and develop a URL shortener service for shortening the given URL.
7. Design and develop a web crawler which uses search engines to discover new or updated content on the web.
8. Design and develop a search autocomplete system which would provide suggestions as you type in the search box.

TOTAL PERIODS

45

COURSE OUTCOMES

On successful completion of this course, students will be able to

CO1 Analyze system requirements for large-scale distributed applications by analyzing real-world problem scenarios. (K4)

CO2 Assess scalable, fault-tolerant, and efficient architectures for distributed systems. (K5)

CO3 Design and develop end-to-end prototypes of distributed systems by considering alternate system design requirements. (K6)

HARDWARE REQUIREMENTS:

- Minimum 8 GB RAM,
- Intel i5 or Ryzen 5 and above

- 256 GB SSD (or HDD if budgeted)

SOFTWARE REQUIREMENTS:

Apache VS Code IDE

Java, JMeter, Postman

MongoDB, Kafka, GitHub, Redis

TEXT BOOKS

1. Alex Xu, Sahn Lam, System Design Interview: An Insider's Guide Volume 2, Open Source: ByteByteGo, 2022
2. Martin Kleppmann, Designing Data-Intensive Applications, O'Reilly Media, Inc, March 2017, (Chapters 5 & 6)

REFERENCE

Github repository:

[GitHub - donnemartin/system-design-primer: Learn how to design large-scale systems.](#)

[Prep for the system design interview. Includes Anki flashcards.](#)

CO-PO-PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2								3		
CO2	3	3	3	3							3	3	
CO3	3	3	3	3	3			3	3		3	3	3